

OpenStax Calculus Volume 2

Section 2.7 Exercises: Integrals, Exponential Functions, and Logarithms

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For the following exercises, find the derivative $\frac{dy}{dx}$.

1. $y = \ln(2x)$
2. $y = \ln(2x + 1)$
3. $y = \frac{1}{\ln x}$

For the following exercises, find the indefinite integral.

4. $\int \frac{dt}{3t}$
5. $\int \frac{dx}{1+x}$

For the following exercises, find the derivative dy/dx . (You can use a calculator to plot the function and the derivative to confirm that it is correct.)

6. [T] $y = \frac{\ln(x)}{x}$
7. [T] $y = x \ln(x)$
8. [T] $y = \log_{10} x$
9. [T] $y = \ln(\sin x)$
10. [T] $y = \ln(\ln x)$
11. [T] $y = 7 \ln(4x)$
12. [T] $y = \ln((4x)^7)$
13. [T] $y = \ln(\tan x)$
14. [T] $y = \ln(\tan(3x))$
15. [T] $y = \ln(\cos^2 x)$

For the following exercises, find the definite or indefinite integral.

16. $\int_0^1 \frac{dx}{3+x}$

17. $\int_0^1 \frac{dt}{3+2t}$
18. $\int_0^2 \frac{x \, dx}{x^2+1}$
19. $\int_0^2 \frac{x^3 \, dx}{x^2+1}$
20. $\int_2^e \frac{dx}{x \ln x}$
21. $\int_2^e \frac{dx}{x (\ln x)^2}$
22. $\int \frac{\cos x \, dx}{\sin x}$
23. $\int_0^{\pi/4} \tan x \, dx$
24. $\int \cot(3x) \, dx$
25. $\int \frac{(\ln x)^2 \, dx}{x}$

For the following exercises, compute dy/dx by differentiating $\ln y$.

26. $y = \sqrt{x^2 + 1}$
27. $y = \sqrt{x^2 + 1} \sqrt{x^2 - 1}$
28. $y = e^{\sin x}$
29. $y = x^{-1/x}$
30. $y = e^{(ex)}$
31. $y = x^e$
32. $y = x^{(ex)}$
33. $y = \sqrt{x} \sqrt[3]{x} \sqrt[6]{x}$
34. $y = x^{-1/\ln x}$
35. $y = e^{-\ln x}$

For the following exercises, evaluate by any method.

36. $\int_5^{10} \frac{dt}{t} - \int_{5x}^{10x} \frac{dt}{t}$
37. $\int_1^{e^\pi} \frac{dx}{x} + \int_{-2}^{-1} \frac{dx}{x}$
38. $\frac{d}{dx} \int_x^1 \frac{dt}{t}$
39. $\frac{d}{dx} \int_x^{x^2} \frac{dt}{t}$
40. $\frac{d}{dx} \ln(\sec x + \tan x)$

For the following exercises, use the function $\ln x$. If you are unable to find intersection points analytically, use a calculator.

41. Find the area of the region enclosed by $x = 1$ and $y = 5$ above $y = \ln x$.
42. [T] Find the arc length of $\ln x$ from $x = 1$ to $x = 2$.

43. Find the area between $\ln x$ and the x -axis from $x = 1$ to $x = 2$.
44. Find the volume of the shape created when rotating this curve from $x = 1$ to $x = 2$ around the x -axis, as pictured here.
[Figure: This figure is a surface. It has been generated by revolving the curve $\ln x$ about the x -axis. The surface is inside of a cube showing it is 3-dimensional.]
45. [T] Find the surface area of the shape created when rotating the curve in the previous exercise from $x = 1$ to $x = 2$ around the x -axis.

If you are unable to find intersection points analytically in the following exercises, use a calculator.

46. Find the area of the hyperbolic quarter-circle enclosed by $x = 2$ and $y = 2$ above $y = 1/x$.
47. [T] Find the arc length of $y = 1/x$ from $x = 1$ to $x = 4$.
48. Find the area under $y = 1/x$ and above the x -axis from $x = 1$ to $x = 4$.

For the following exercises, verify the derivatives and antiderivatives.

49. $\frac{d}{dx} \ln(x + \sqrt{x^2 + 1}) = \frac{1}{\sqrt{1+x^2}}$
50. $\frac{d}{dx} \ln\left(\frac{x-a}{x+a}\right) = \frac{2a}{(x^2-a^2)}$
51. $\frac{d}{dx} \ln\left(\frac{1+\sqrt{1-x^2}}{x}\right) = -\frac{1}{x\sqrt{1-x^2}}$
52. $\frac{d}{dx} \ln(x + \sqrt{x^2 - a^2}) = \frac{1}{\sqrt{x^2-a^2}}$
53. $\int \frac{dx}{x \ln(x) \ln(\ln x)} = \ln(\ln(\ln x)) + C$